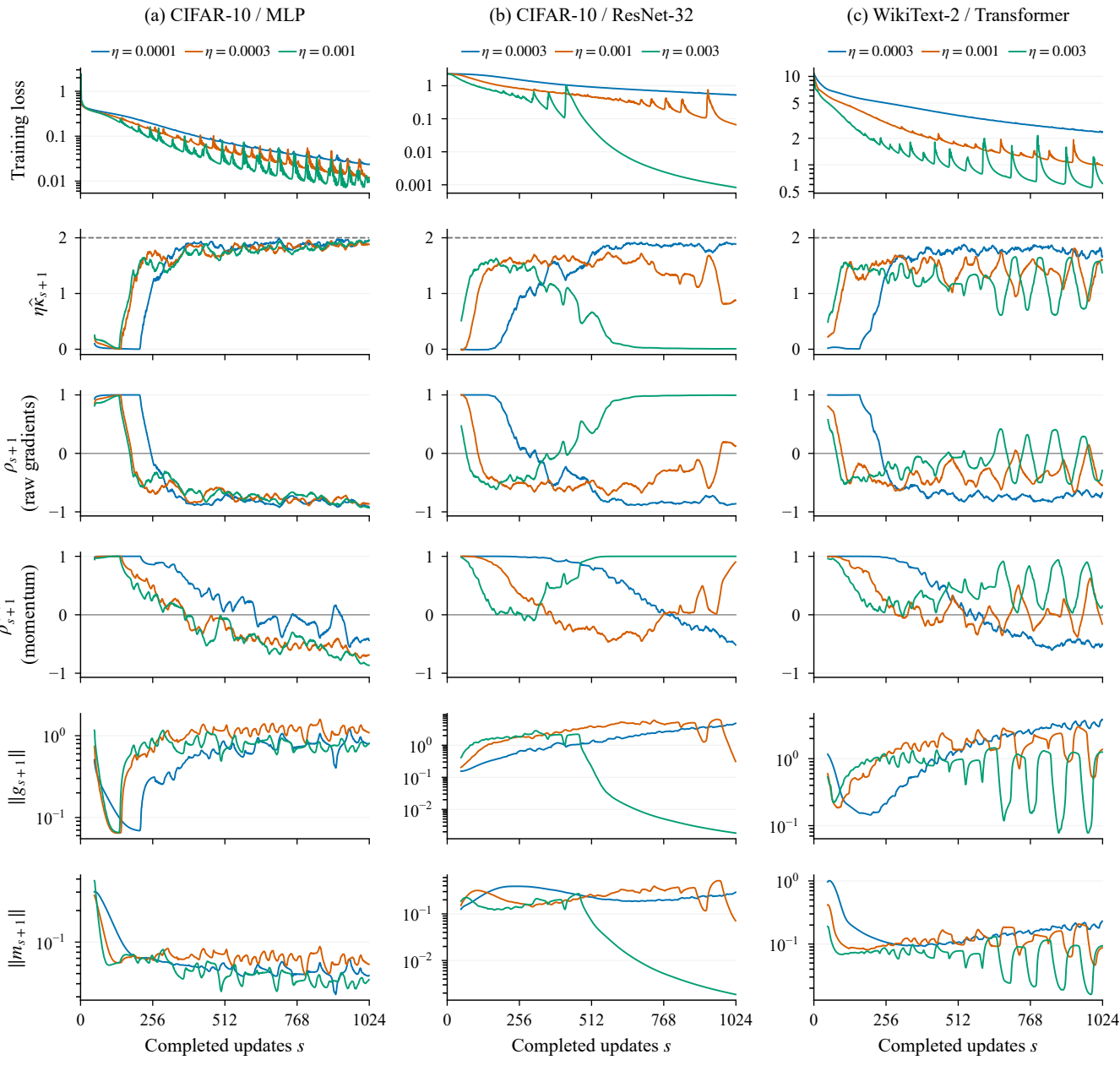

(a) CIFAR-10 / MLP
(b) CIFAR-10 / ResNet-32
(c) WikiText-2 / Transformer
η = 0.0001
η = 0.0003
η = 0.001
η = 0.003
Training loss
$\hat{\eta}\hat{K}_{s+1}$
ρ_{s+1} (raw gradients)
$\rho^{(m)}_{s+1}$ (momentum)
$\|g_{s+1}\|$
$\|m_{s+1}\|$
Completed updates s